

Where are you? Influence of Redirected Walking on Audio-Visual Position Estimation of Co-Located Users

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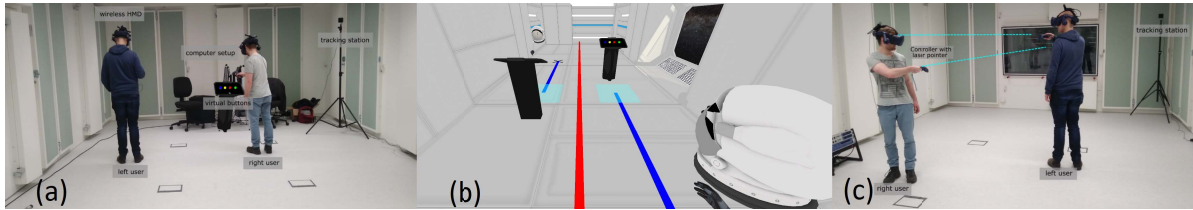


Figure 1: Co-located multi-user redirection: (a) two users during the study with consoles, (b) view of the virtual environment with two avatar representations walking towards each other, (c) users pointing towards each other.

ABSTRACT

Two-user redirected walking (RDW) promises new possibilities for collaborative experiences in virtual reality (VR), but it also introduces challenges like the spatial de-synchronization emerging when one or more users are redirected in different ways. We analyzed the ability to estimate a co-located user's position, and assessed the feeling of representation consistency in an interactive game. Results show that a user's estimation of the other's position improves when their partner's virtual and physical position get closer together. Furthermore, larger redirection gains lead to a lower feeling of *consistency*, which also applies if only one person is redirected.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

RDW imperceptibly manipulates the user's position or rotation, which results in users moving on different paths in VR compared to the paths they are walking on in the real world [4]. When using multi-user RDW, several co-located users have spatial relations with respect to each other in the virtual environment (VE) which might vary from those in the real world [2] (i.e., spatial de-synchronization [1]). In co-located, but not collaborative, scenarios, researchers have found that users are generally not aware of another person in the room, especially if they have not been informed that there might be someone in the same room [3]. However, the perceptual effects of the de-synchronization between users in a co-located and collaborative VR scenario, where users interact with each other, have not been studied in sufficient detail yet. For instance, it is unknown where users perceive the position of another co-located user, when the real voice comes from a different location, or if users can detect the redirection of a co-located person. We developed an interactive VR game, where two users are redirected, resulting in different spatial relations in the real and virtual world. We measured pointing errors towards the estimated physical position of the other user, as well as the users' feeling of *consistency*, which we define as the perception of a consistent audio-visual representation of others.

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Additionally, we recorded whether the users thought that they or their partner were redirected.

2 STUDY

In this study, two participants performed a collaborative task in a co-located environment. 34 users (19 male, 15 female, age = 28.88 years $\pm$ 10.56 years) took part in the study in pairs of two. It took approximately 45 minutes, with around 25 minutes in VR while users wore head-mounted displays (HMD). The experimental software was running on two computers to enable the co-located experience (PC1: i7-4930K, NVIDIA GeForce RTX 2080 Ti; PC2: i7-5930K, NVIDIA GeForce GTX 980). As immersive HMDs we used two HTC Vive Pro with VIVE Wireless Adapters.

Both users were located in the same physical room and connected to the same VE via network. They could see a synchronized virtual representation of their partner's head and hand, and directly hear their voice. Their task was to walk towards one of two consoles. There, one user had four buttons and the other one could see a color code (see Figure 1). They had to communicate the code by talking to each other, and type it in correctly by touching the corresponding buttons. The communication between users provided auditory cues about their positions for the position judgment task. Moreover, this gamified scenario mimicked a realistic multi-user experience. While walking, the users were redirected. There were twelve possible conditions, which were shuffled randomly (See Figure 2). The following gains were used: $G_{T1} = 1$, $G_{T2L} = 1.2$, $G_{T3R} = 0.8$, $G_{T4} = 1$, $G_{T5} = 1.25$, $G_{T6} = 0.8$, $G_{C1-C3} = 2^\circ/m$, $G_{C4-C6} = 5^\circ/m$

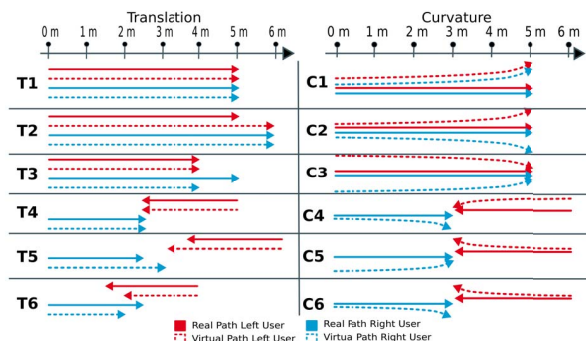


Figure 2: All conditions in the study.

Once they typed in the code correctly, the users' avatars were turned off, the virtual room was replaced by a slightly larger empty room and the three following questions were asked:

1. Were you or your partner redirected? (Me, My Partner)
2. How consistent was the experience with your partner? (How much did what you see coincide with what you heard and felt, on a scale from 0 to 3?)
3. Please point to where your partner is located physically. (With a laser pointer attached to the controller.)

When both users were done, they walked to their new starting position, marked with a cross on the ground and the next trial started.

3 PRELIMINARY RESULTS

This section provides an overview over the results we found. Regarding question 1, the majority of participants thought their partner was redirected in the translation conditions, even when they were the only one whose position was altered. Only those users with a gain of 0.8 (T_3 Left and T_6) detected their own redirection more often than not. Curvature was easier to detect, especially the three conditions with $5^\circ/\text{m}$.

The results for question 2 can be seen in Figure 3. For translation, the two conditions without redirection received the highest rating with means of 2.47 ± 0.61 for T_1 and 2.35 ± 0.65 for T_4 . For curvature, C_1 receive the highest rating with 2.41 ± 0.7 . There is no significant difference between the left and right user in T_2 (left: 2.29 ± 0.59 , right: 1.88 ± 0.93 , $p = 0.187$) and T_3 (left: 1.82 ± 0.95 , right: 1.76 ± 0.97 , $p = 0.856$), where only one user was redirected.

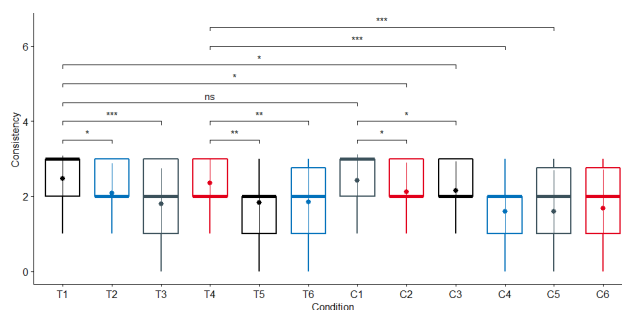


Figure 3: Boxplots with mean (dot) and median (thick line) for *consistency*, on a scale from 0 to 3. Important statistical differences are highlighted (Wilcoxon Signed-Rank Test, 0.05 sig. level).

The results for question 3 can be seen in Figure 4. The lowest pointing errors could be found for T_1 ($6.49^\circ \pm 4.72^\circ$) and for C_1 ($5.77^\circ \pm 4.83^\circ$). The conditions with the highest pointing errors are T_2 ($14.08^\circ \pm 12.15^\circ$), T_3 ($14.83^\circ \pm 12.17^\circ$) and T_6 ($14.03^\circ \pm 11.20^\circ$).

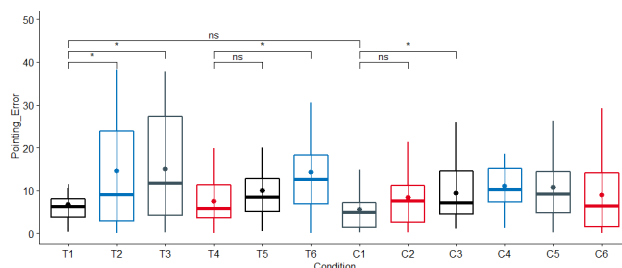


Figure 4: Boxplots with mean (dot) and median (thick line) for the pointing error. Important statistical differences are highlighted (Wilcoxon Signed-Rank Test, 0.05 sig. level).

4 DISCUSSION

Regarding question 1, an interesting aspect is the fact that some users suspected that they were redirected, even though their partner was. Hence, the redirection of their partner also influenced them to think that they were redirected. Therefore, it is important to not only consider single-user redirected walking thresholds when redirection should be undetected, but also to keep the spatial relation and distance between multiple user as consistent as possible.

For question 2, *consistency* was lower when stronger redirection gains were used, and it is not significantly different for the left and right user in T_2 and T_3 , even though only one of them was redirected. This indicates that even if users were not actively redirected, their feeling of *consistency* was influenced by their partner's redirection. The study produced the lowest pointing errors for question 3 when the distance between the other participant's physical and virtual position was smaller, e.g. in T_1 , T_4 and C_1 . The results indicate that VR users still highly rely on their visuals, and neither auditory cues nor the physical presence of another person can fully overwrite this information.

This should be noted especially in multi-user experiences, where the discrepancy can lead to a lower feeling of consistency and impair their experience. In addition to being detrimental to their experience, being far away virtually, but close physically can lead to collisions and accidents, because the users cannot fully assess their partner's real position.

On the other hand, the domination of visuals also shows that multi-user RDW is possible, because the perceived discrepancy between the virtual and physical position is smaller than it really is. With correct adjustments of the game mechanisms, it is possible to implement realistic multi-user RDW applications, especially if the users' safety is guaranteed.

5 FUTURE WORK

This poster provides first insight into the detection of a partner's physical position during a co-located collaborative VR experience when RDW is applied. The presented results will be further evaluated to gain more insights. In future studies, we plan to test how different distances between the virtual users influence their experience, e.g. if larger distances or very small distance have a stronger effect on their position estimation, feeling of consistency or ability to detect redirection. In addition, the pointing error should also be investigated in more detail. We recorded only the absolute angle between the pointing direction and the position of the other user. Alternatively, a two-alternative forced choice test might be used by asking users if their partner is walking faster or slower virtually, compared to the physical motion.

Future research can use the information from this study to design applications, keeping in mind the influence of redirected walking on redirection detection, *consistency* and the localization of co-located users in collaborative virtual environments.

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